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# Contrastive Vision-Language Learning

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# Contrastive Vision-Language Learning

## 01 / Motivation

### The Bottleneck of Supervision

Traditional computer vision models rely heavily on expensive, manually annotated datasets with fixed categories.

This strict reliance severely limits model scalability and adaptability to open-ended, real-world visual concepts.

## 02 / Main Idea

### Shared Latent Alignment

Contrastive vision-language learning bridges this gap by training dual encoders.

It directly aligns images and raw natural language captions within a unified, shared latent representation space.

>76%

Zero-Shot Accuracy

## 03 / Key Results

Models like **CLIP** demonstrate unprecedented zero-shot transfer capabilities across benchmarks.

Trained on hundreds of millions of image-text pairs, achieving >76% **ImageNet accuracy** without fine-tuning.

Revolutionizes open-vocabulary classification.

# Evolution of Contrastive Visual Representations

## Contrastive Learning Foundations

Methods like **SimCLR** and **MoCo** established the viability of self-supervised learning using [InfoNCE](#) loss to pull positive samples together and push negative samples apart.

## Vision-Language Evolution

Early approaches like **ConVIRT** successfully applied contrastive alignment to medical imaging. This paved the way for massive-scale foundational models, shifting the paradigm from unimodal representations to robust, multimodal feature alignment.

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M**

**OpenAI CLIP**  
Curated Image-Text Pairs

**1B**

**Google ALIGN**  
Scaled Noisy Pairs

# System Architecture & Training Pipeline

## 01

### Dual-Encoder Architecture

The system utilizes a **dual-encoder design** to process multimodal inputs asynchronously.

#### Image Encoder:

- Vision Transformer (ViT)
- ResNet Variants

#### Text Encoder:

- Standard Transformer

## 02

### Contrastive Objective

Optimized using a **symmetric cross-entropy contrastive loss**.

#### Key Mechanism:

- **Maximizes** cosine similarity for matching image-text pairs.
- **Minimizes** cosine similarity for all incorrect pairings within the batch.

## 03

### Large-Batch Training

Success relies heavily on scaled mini-batch optimization.

**Batch Size: Up to 32,768**

Provides a sufficient number of **negative samples** per iteration.

Ensures robust and **highly discriminative** feature learning.

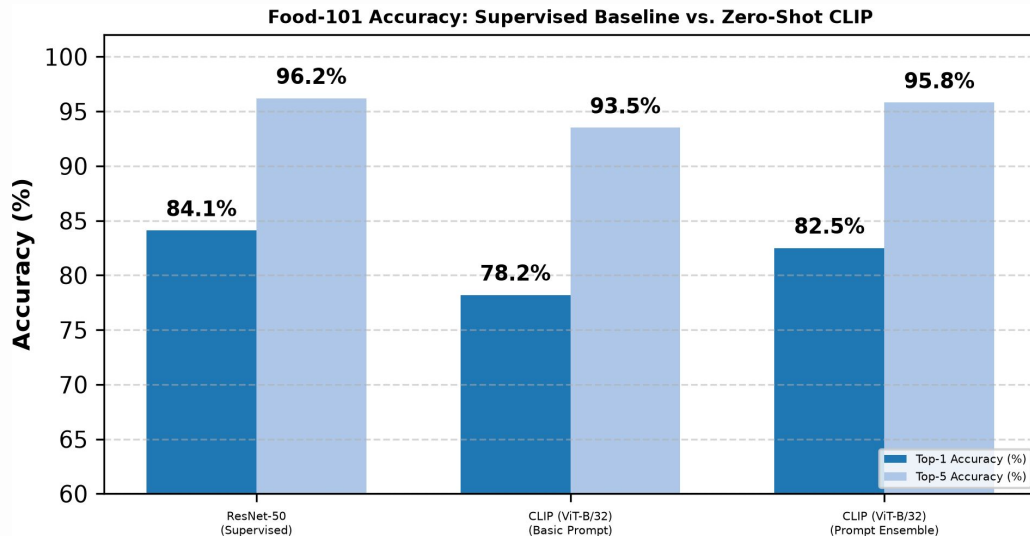
## Zero-Shot Evaluation Result

# Food-101 Benchmark: CLIP vs. Fully Supervised Model

**Prompt Ensemble Significantly Improves Performance**

Using Prompt Ensemble improved Top-1 accuracy from **78.2%** to **82.5%** (+4.3%).

This demonstrates that multiple prompt word designs effectively reduce semantic ambiguity and significantly unlock the zero-sample potential of CLIP.



Zero-sample performance approaches the fully supervised benchmark.

CLIP achieves a Top-5 accuracy of **95.8%** without any fine-tuning using the Food-101 data.

It is extremely close to the ResNet-50 fully supervised benchmark (96.2%), demonstrating strong cross-domain open-vocabulary generalization ability.

# Project Repository & Setup



## Repository Access

Access the complete codebase, model weights, and documentation at:

[<https://github.com/simon3240/Contrastive-Vision-Language-Learning/tree/main>]

## Implementation Steps

- 1 Repository Setup:** Clone the repository and install dependencies via the provided [requirements.txt](#) file.
- 2 Data Pipeline:** Dataset  
<https://huggingface.co/datasets/ethz/food101>
- 3 Model Training:** Execute the main training script to begin dual-encoder alignment.

## Evaluation & Analysis

The repository includes interactive **Jupyter notebooks** designed for end-to-end model inspection:

- Evaluating zero-shot classification performance
- Visualizing the learned multimodal embedding space